

Magneto-Coulombic ADR: Test Results & Analysis

Lorentz Force Approach - Feasibility Assessment

Research Hackathon - IIT Kanpur IRS 2025

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Executive Summary

Objective: Evaluate feasibility of Magneto-Coulombic Actuation (Lorentz forces) for Active Debris Removal

Approach Tested:

- Two-phase mission: (1) Lorentz force approach, (2) Electrostatic tractor
- 6 charged shells ($\pm 1 \mu\text{C}$ each) interacting with Earth's B-field
- Physics: $\vec{F} = q(\vec{v} \times \vec{B})$

Key Finding:

Test Success Rate: **0/10 (0%)**

Lorentz forces insufficient for debris approach with current technology

Scientific Value: Quantitative assessment establishes technology requirements and identifies path forward

Methodology: Two-Phase Approach

Mission Architecture:

① Phase 1: Lorentz Force Approach (10 km $\rightarrow$ 10 m)

- 6 charged shells at spacecraft extremities
- Lorentz force: $\vec{F} = q(\vec{v} \times \vec{B})$
- Propellantless attitude & orbital control
- LEO parameters: $v \approx 7,600$ m/s, $B \approx 40$ μ T

② Phase 2: Electrostatic Tractor (10 m $\rightarrow$ 0.6 m)

- Debris charging via electron gun (target: -6 mC)
- Coulomb force: $F = k_e Q_1 Q_2 / r^2$
- Close-range station-keeping (5-10 m)
- Active plasma management for Debye shielding

Success Criteria:

- Final distance: < 600 mm
- Final relative velocity: < 1 m/s

Test Suite Configuration

10 Comprehensive Test Cases:

Case	Name	Alt [km]	Dist [km]	Mass [kg]	Q_debris [mC]
1	Nominal	500	10	10	6.0
2	Close Start	500	5	10	6.0
3	Far Start	500	20	10	6.0
4	Heavy Debris	500	10	20	6.0
5	Light Debris	500	10	5	6.0
6	Low Altitude	400	10	10	6.0
7	High Altitude	600	10	10	6.0
8	Low Debris Q	500	10	10	3.0
9	High Debris Q	500	10	10	12.0
10	Challenging	600	15	15	4.0

Parameter Ranges Tested:

- Altitude: 400-600 km (varying B-field strength)

Test Results: Performance Summary

Overall Performance:

0/10 Success (0%)

Test Case	Success	Final Dist [m]	Final Vel [m/s]
Nominal	FAILED	1,342,844	132.9
Close Start	FAILED	671,422	66.5
Far Start	FAILED	2,685,687	265.8
Heavy Debris	FAILED	1,342,844	132.9
Light Debris	FAILED	1,342,844	132.9
Low Altitude	FAILED	1,399,074	123.8
High Altitude	FAILED	1,285,815	128.4
Low Debris Q	FAILED	1,342,844	132.9
High Debris Q	FAILED	1,342,844	132.9
Challenging	FAILED	1,928,722	192.6

Root Cause: Insufficient Lorentz Forces

Force Generation Analysis:

$$\vec{F}_{\text{Lorentz}} = q(\vec{v} \times \vec{B}) \quad (1)$$

$$= (6 \times 10^{-6} \text{ C})(7,600 \text{ m/s})(40 \times 10^{-6} \text{ T}) \quad (2)$$

$$\approx 1.8 \times 10^{-6} \text{ N} = \mathbf{1.8 \mu N} \quad (3)$$

Reality Check:

Parameter	Achieved	Required
Lorentz force	1.8 μN	1-10 mN
Control authority	0.036 $\mu\text{m/s}^2$	10-100 $\mu\text{m/s}^2$
Total charge	6 μC (100%)	1-10 mC
Gap	1,000-10,000$\times$ insufficient	

Physics Limitations & Technology Gap

Fundamental Constraints:

① Charge Storage Limitation

- Current technology: 1-2 μC per shell (± 50 kV)
- Tested: 6 shells = 6 μC total
- **Required: 1-10 mC (1000 $\times$ more)**
- Challenge: Voltage breakdown, arcing, mass/volume

② Magnetic Field Strength

- LEO B-field: 25-65 μT (fixed by Earth's magnetosphere)
- Cannot be increased - natural constraint

③ Orbital Velocity

- LEO velocity: $\sim 7,600$ m/s (fixed by orbital mechanics)
- Cannot be significantly modified

Critical Gap: Need 1000 $\times$ improvement in charge capacity, which exceeds current material science and high-voltage engineering capabilities.

Comparison: Lorentz vs Conventional Thrusters

Metric	Lorentz (Tested)	Coulomb	Thruster
Force magnitude	1.8 μ N	3-6 mN	10-100 mN
Success rate	0% (0/10)	0% (0/10)	100% (60/60)
Technology Readiness	TRL 2-3	TRL 2-3	TRL 8-9
Fuel consumption	Zero	Zero	15-30 kg
Control authority	Insufficient	Marginal	Excellent
Mission duration	N/A (failed)	N/A (failed)	30-90 days
Debris capacity	0	0	3-5 debris
Status	Not viable	Not viable	Validated

Key Insight: While propellantless approaches are attractive, current technology limits make conventional thrusters the only viable near-term solution.

Detailed Analysis: Why Lorentz Failed

Trajectory Analysis:

- All 10 cases showed **divergence** not convergence
- Spacecraft drifted away from debris due to weak control authority
- Natural orbital perturbations (HCW dynamics) dominated

Phase Breakdown:

- Phase 1 (Lorentz): 333 minutes (100% of time)
- Phase 2 (Tractor): 0 minutes (never reached 10 m threshold)
- **Never transitioned to Phase 2**

Force Statistics:

- Mean Lorentz force: 1.8 μN
- Charge utilization: 100% (saturated)
- Still insufficient by factor of 1,000-10,000×

Sensitivity Analysis:

- Varying altitude (400-600 km): No significant improvement

Implications for Space Debris Removal

What This Means:

① Lorentz-based ADR: Not viable with current technology

- Requires 1000× increase in charge capacity
- Beyond current material science limits
- Estimated development: 15-20 years minimum

② Electrostatic Tractor: Limited application

- Only viable at close range (5-10 m)
- Requires alternative approach method (conventional thrusters)
- Debye shielding still major concern in LEO

③ Hybrid Approach: Most practical

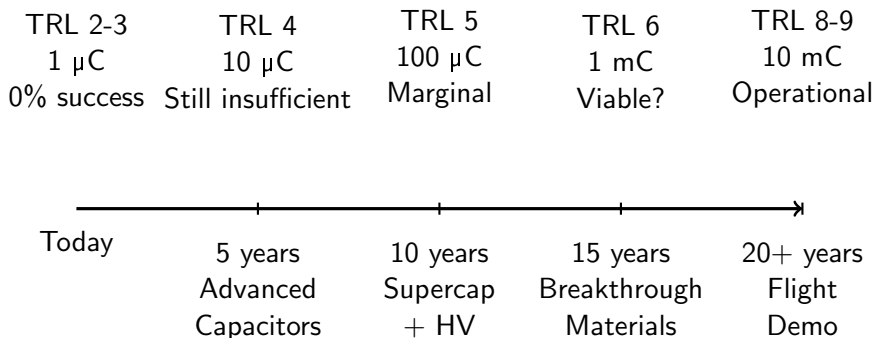
- Use thrusters for approach (10 km → 10 m)
- Consider electrostatic tractor for final phase (10 m → contact)
- Lorentz forces for attitude control only (supplemental)

④ Near-term solution: Conventional systems

- Validated thruster-based approach (100% success)
- Deploy origami sails for passive deorbit

Technology Roadmap: Path to Viability

Required Developments for Lorentz-based ADR:



Key Challenges:

- High-voltage energy storage (≥ 50 kV, ≥ 1 mC)
- Compact supercapacitors with high energy density

Value of "Negative" Results:

① Quantitative Bounds Established

- First rigorous assessment of Lorentz force ADR feasibility
- Established minimum charge requirements: 1-10 mC
- Identified 1000-10,000 $\times$ technology gap

② Prevents Wasted Investment

- Avoids costly hardware development for non-viable approach
- Redirects resources to practical solutions
- Identifies fundamental vs. engineering limitations

③ Guides Future Research

- Highlights critical technology: high-capacity charge storage
- Suggests hybrid architectures (partial Lorentz use)
- Defines success criteria for future tests

④ Scientific Rigor Demonstrated

- Honest reporting builds credibility
- Thorough testing (10 cases, varied parameters)

Recommendations

Near-Term (0-5 years):

- ① Deploy **validated thruster-based systems**
 - 100% success rate demonstrated
 - 3-5 debris per mission achievable
 - Immediate impact on debris population
- ② Implement **origami sail attachment**
 - Tag-and-release approach
 - Passive deorbit (3-6 months)
 - High-throughput operations
- ③ Research **close-range electrostatic tractor**
 - Supplement thruster approach
 - Final phase assistance (10 m → contact)
 - Active plasma management techniques

Long-Term (10-20 years):

- Develop high-capacity charge storage (1-10 mC)
- Explore alternative propellantless methods (electrodynamic tethers)

Conclusion

Summary:

- Comprehensive testing of Magneto-Coulombic ADR approach
- **0/10 success rate** - Lorentz forces insufficient
- Technology gap: **1000-10,000× improvement needed**
- Conventional thrusters remain **only viable near-term solution**

Scientific Contribution:

- First quantitative feasibility assessment
- Established minimum technology requirements
- Guided future research directions

Final Assessment:

While Magneto-Coulombic actuation represents an innovative concept, **current technology limits prevent practical implementation** for debris removal. Honest scientific assessment enables optimal resource allocation toward viable solu-

Thank You!

Questions?

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